

Intent Ledger & Intentional Undo

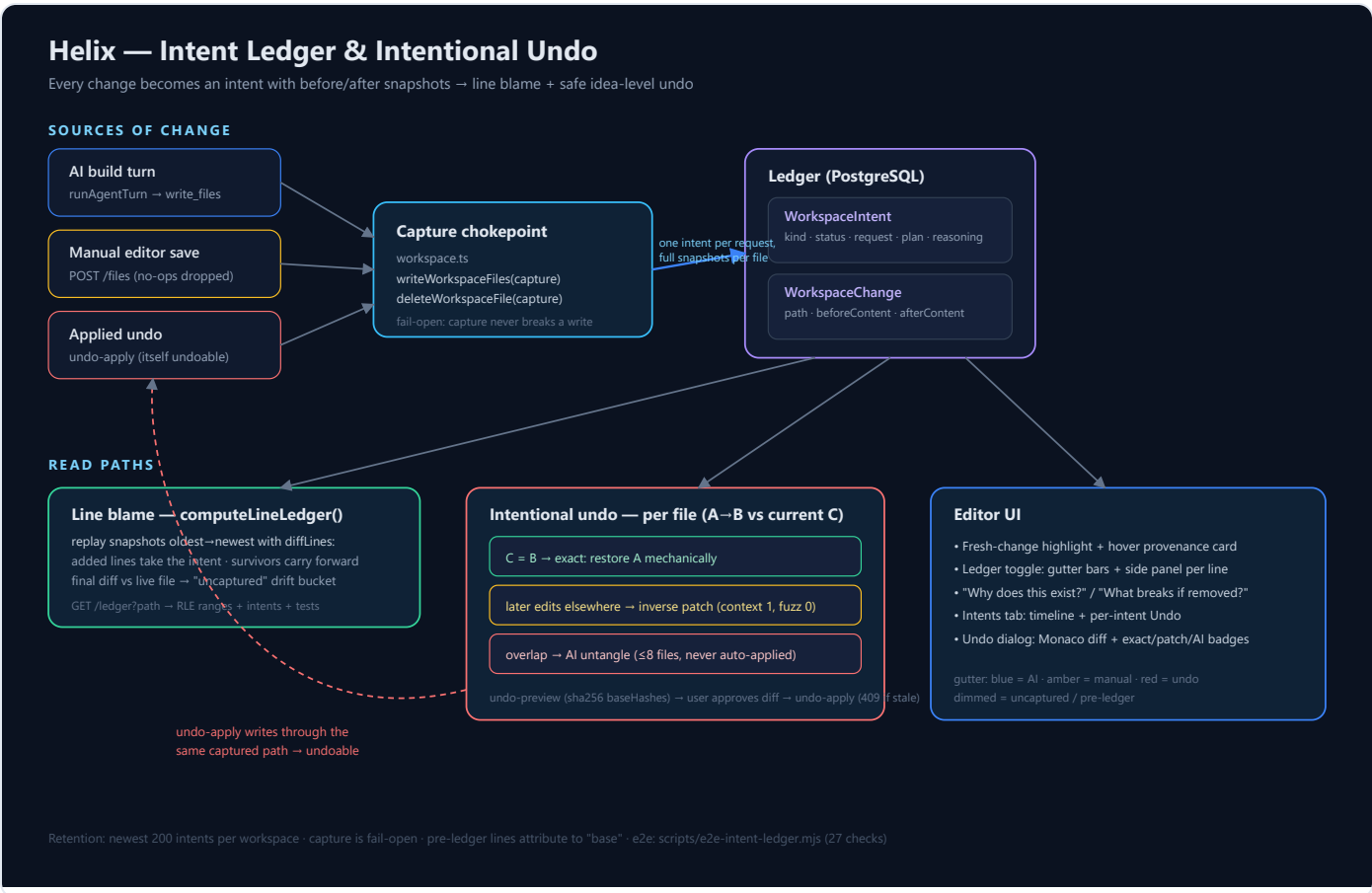
Helix Studio · shipped 2026-06-12 · lives in the editor (`/editor/[id]`)

What it is

Two connected capabilities that give every line of code in a Helix workspace a memory:

- **Line-By-Line Intent Ledger** — every change (an AI build turn, a manual editor save, or an applied undo) is recorded as an *intent* with full before/after snapshots of each file it touched. Any line in the editor can answer: *who introduced me, from which request, implementing which plan step, protected by which tests?*
- **Intentional Undo** — revert an *idea*, not a commit. "Remove the invite feature" reverses every file that intent introduced — code, new files, deletions — while preserving all later work, behind a preview diff the user approves.

Architecture



WorkspaceChange	One per file an intent touched	path , beforeContent , afterContent (full snapshots; null = absent/deleted), baseUnknown
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Why full snapshots, not diffs: files are capped at 48 KB so the worst case is bounded; exact-restore becomes a string compare; and a missing link can't break a replay chain — drift from uncaptured writes self-heals into an "uncaptured" blame bucket. Retention keeps the newest **200 intents per workspace**.

Capture (write chokepoint)

Every write funnels through `writeWorkspaceFiles()` / `deleteWorkspaceFile()` in `src/lib/workspace.ts`, which accept an optional `capture: {intentId}`:

- **AI turns** — `runAgentTurn()` creates the intent lazily on the first mutating tool call (read-only turns leave no rows) and finalizes it with the assistant's reply as `reasoning`; approved plans are stored as `planText`. Verify auto-fix turns fold into the parent intent.
- **Manual saves** — the files route drops no-op saves, then records one `manual` intent.
- **Undo** — `applyUndo()` writes through the same captured path, so an undo is itself an intent — and therefore undoable.

Capture is **fail-open**: if snapshotting fails, the write still succeeds and blame self-heals later.

Blame (line → intent)

`computeLineLedger()` replays a file's snapshots oldest-first with jsdiff `diffLines`: added lines take that change's intent, surviving lines carry attribution forward, and a final diff against live content buckets unexplained lines as *uncaptured*. Output: run-length-encoded ranges + intent metadata + a protecting-tests heuristic. Computed only when the ledger UI asks — never per keystroke.

Undo engine

Case (A = before, B = after, C = current)	Method
C === B — untouched since the intent	<code>exact</code> restore A mechanically
Later edits elsewhere in the file	<code>patch</code> inverse patch B→A applied to C (jsdiff, context 1, fuzz 0 — a context mismatch is overlap)
Overlapping edits / created-then-edited / deleted-then-recreated / unknown base	<code>ai</code> one no-tools model call produces the post-revert file (≤8 files per preview, never auto-applied)

Preview/apply protocol: `undo-preview` returns proposed entries + sha256 `baseHashes` of live content and never writes; `undo-apply` re-verifies the hashes (workspace moved on → **409**, client re-previews), records the revert as a new `undo` intent, and marks the target `reverted`.

API routes

Route	Returns
GET /api/workspaces/[id]/ledger?path=	line ranges → intents, protecting tests
GET /api/workspaces/[id]/intents	newest-first intent timeline
POST /api/workspaces/[id]/ledger/ask	grounded AI answer to "why" / "impact" (token-metered)

POST .../intents/[intentId]/undo-preview	proposal: entries, unresolved, baseHashes
POST .../intents/[intentId]/undo-apply	{ changes } , or 409 on stale preview

UI

- **Fresh-change highlight (always on):** lines the latest turn introduced get a soft accent background; hovering shows a compact provenance card next to the block (click pins) — title, when, a short why, *Why? / If removed?* AI buttons, and Undo.
- **Ledger toggle:** colored gutter bars (blue = AI, amber = manual, red = undo, dimmed = uncaptured) + a side panel for any clicked line.
- **Intents tab:** the change timeline; per-intent Undo opens a preview dialog with exact/patched/AI badges and a Monaco side-by-side diff.
- The Save button reads "**Saved ✓**" when nothing is dirty — AI edits persist automatically; Push ships to the repo.

Limits & verification

- Lines written before the feature shipped (and imported-repo base lines) show "predates the ledger".
- AI-untangled revert files always carry an explicit **AI** badge and deserve review before applying.
- End-to-end test: `node scripts/e2e-intent-ledger.mjs` — 27 checks covering capture → blame → undo → undo-the-undo → stale-preview 409.